Lecture 12: Meteorites, Asteroids, Minor Planets & Comets

Planetary Systems

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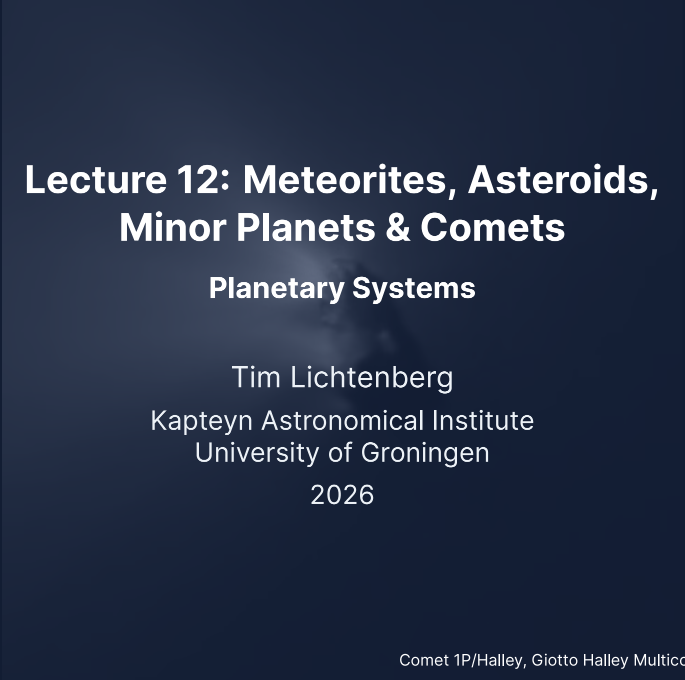


Comet 1P/Halley, Giotto Halley Multicolour Camera (1986). Credit: ESA

Learning Objectives

By the end of this lecture, you will be able to:

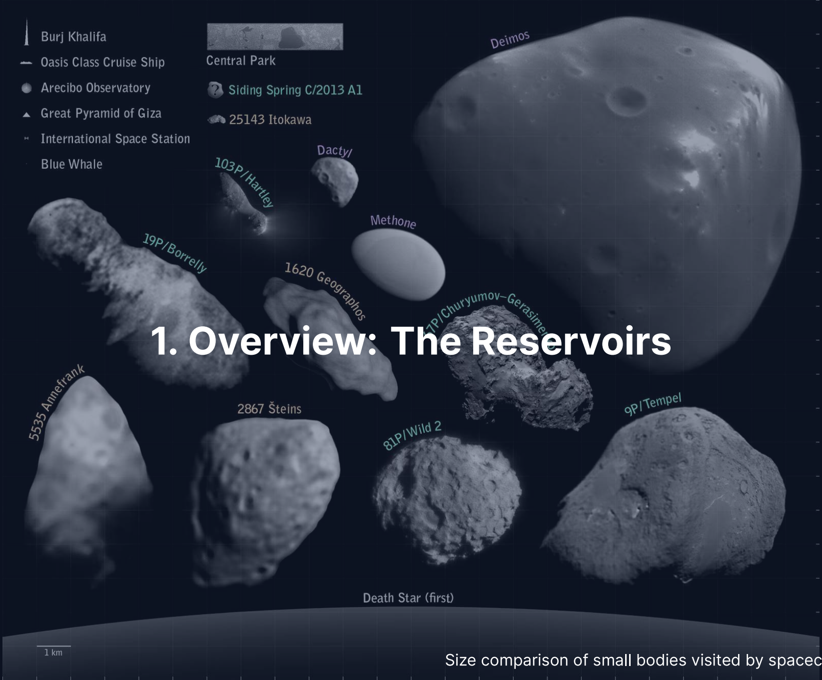
- Classify meteorites and use isotopic chronometers to date the early solar system
- Describe the asteroid-belt and trans-Neptunian populations and their dynamics
- Explain the origin, anatomy and volatile composition of comets
- Derive the Pb-Pb isochron age of calcium-aluminium-rich inclusions (CAIs)
- Read small bodies as fossils of planet formation

Recap: Where We Are in the Course

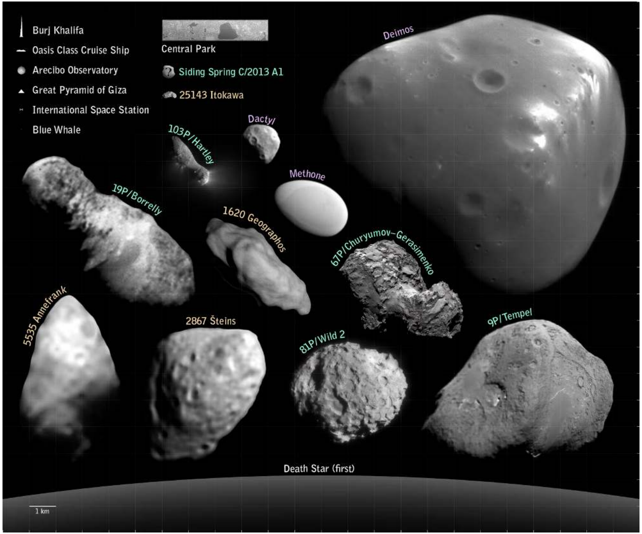
- Lectures 9–11 surveyed the major planets across the rocky and giant classes
- Today: the leftovers; asteroids, meteorites, KBOs, comets, and the Oort cloud
- Small bodies act as formation fossils preserving disk chemistry and migration history

Today: What do the leftovers reveal about the origin of planetary systems?

1. Overview: The Reservoirs



Size comparison of small bodies visited by spacecraft. NASA/JPL-Caltech



Small bodies visited by spacecraft, scaled against terrestrial landmarks; three reservoirs hold these bodies: the main belt (2.1–3.3 AU, ~ $4 \times 10^{-4}\ M_{\oplus}$), the Kuiper Belt, and the Oort cloud (2,000–50,000 AU, inferred, never directly observed). Credit: NASA/JPL-Caltech, public domain.

2. Meteorite Taxonomy

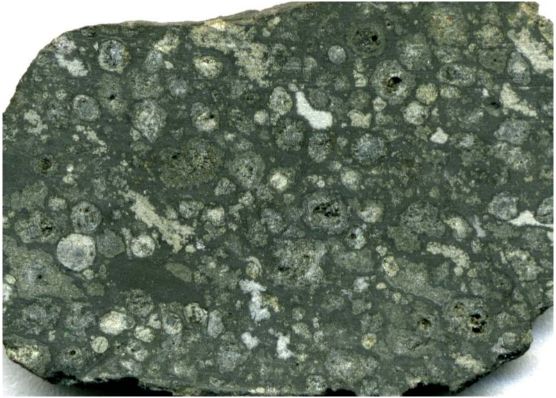
Allende meteorite (CV3 chondrite, fell 1969). J. St. John CC BY 2.0

Why Meteorites Matter

Meteorites are the oldest rocks accessible in terrestrial laboratories, preserving disk-condensation grains with unparalleled chronological precision:

- **Laboratory access:** ~70,000 catalogued specimens, older than the oldest terrestrial zircons (~ 4.4 Ga)
- **Presolar material:** primitive chondrites contain grains that condensed around other stars, older than the Sun
- **Anchor age:** CAI crystallization age of 4567.30 ± 0.16 Myr anchors solar system history

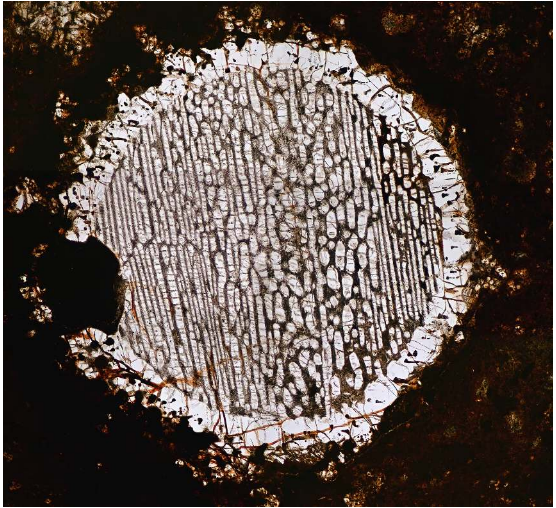
The First Cut: Chondrites vs. Everything Else



Allende slab: chondrules + bright CAIs in dark matrix

- **Chondrites**: never melted whole. Aggregates of disk solids.
- Characteristic: ~mm-scale silicate *chondrules*.
- Achondrites, irons, stony irons: from *differentiated* parent bodies.
- Differentiation needs heat → short-lived $^{26}\mathrm{Al}$.
- Iron meteorites = cores; achondrites = crusts/mantles.

Chondrules in Thin Section

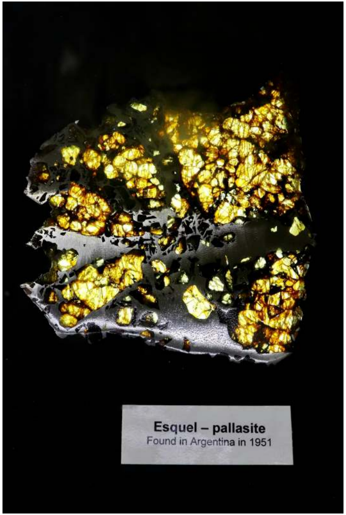


- Cross-polarised thin section, NWA 5930.
- Each chondrule: ~mm bead of olivine + pyroxene.
- Frozen from a fully molten droplet in seconds to hours.
- Peak T ~ 1500–1900 K.
- How chondrules form remains an unsolved problem.

Iron Meteorites: Reading a Cooling Rate

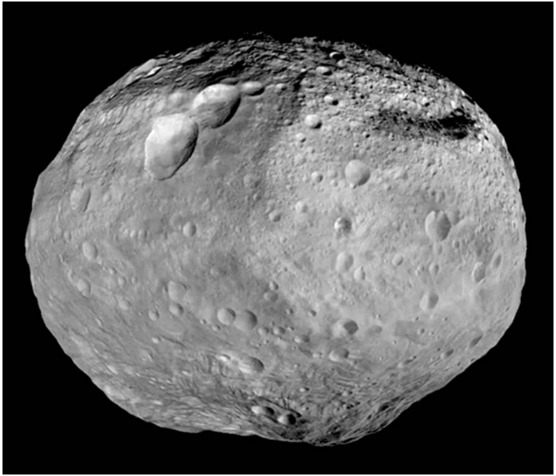
- Widmanstätten pattern: kamacite + taenite Fe-Ni lattice.
- Forms only at cooling rates $\lesssim 100$ K Myr^{-1}.
- Slow cooling possible only inside metallic cores of ~tens-of-km bodies.
- Lattice spacing → original parent-body radius.

Pallasite: A Core-Mantle Boundary in Hand



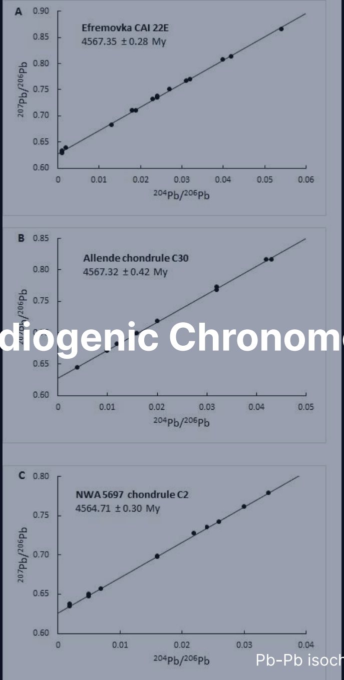
- Esquel pallasite: yellow-green olivine in silvery Fe-Ni metal.
- Texture suggests boundary between molten core and silicate mantle.
- Olivine crystals sank into metal during a giant impact.
- **A direct sample of a destroyed planetesimal's core-mantle boundary.**

Vesta: HED Parent Body



- HED meteorites (howardites/eucrites/diogenites) = ~ 6% of falls.
- Identical mineralogy + oxygen isotopes.
- Dawn (2011–2012): confirmed Vesta as parent.
- Rheasilvia basin (south pole): excavated and ejected the HEDs.
- Lunar + Martian meteorites identified the same way (oxygen isotopes + trapped Mars atmosphere).

3. Radiogenic Chronometers



Pb-Pb isochrons for CAI 22E and chondrules. Connelly et al. (2012)

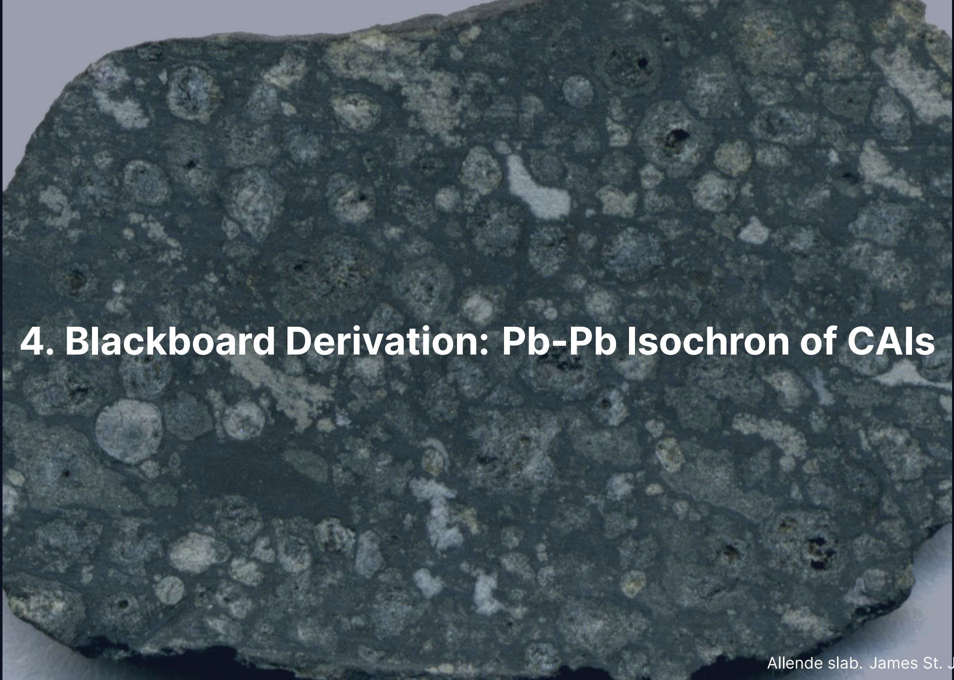
Decay Clocks and Early Solar System Timeline

Radioactive decay $\frac{dN}{dt} = -\lambda N$ ($t_{1/2} = \ln 2/\lambda$) provides absolute and relative chronometers:

- **Long-lived:** $^{238}\mathrm{U}$, $^{235}\mathrm{U}$ yield absolute ages; CAIs condense at $t_0 = 4567.30 \pm 0.16$ Myr
- **Short-lived (extinct):** $^{26}\mathrm{Al}$ (0.717 Myr), $^{182}\mathrm{Hf}$ (8.9 Myr) yield relative ages
- **Early chronology:** Differentiation at 0.5–4 Myr, chondrules at 1–4 Myr after CAIs

Long-lived chronometers anchor absolute time; extinct nuclides resolve events to < 1 Myr.

4. Blackboard Derivation: Pb-Pb Isochron of CAIs



Setup: Two Parallel Chains in One Element

- ${}^{238}\mathrm{U} \to {}^{206}\mathrm{Pb}$ with $\lambda_{238} = \ln 2/(4.4683\,\mathrm{Gyr})$.
- ${}^{235}\mathrm{U} \to {}^{207}\mathrm{Pb}$ with $\lambda_{235} = \ln 2/(0.7038\,\mathrm{Gyr})$.
- ${}^{204}\mathrm{Pb}$: stable, no significant radiogenic source. The reference isotope.

Radiogenic daughter at time t since closure:

$$ {}^{206}\mathrm{Pb}^* = {}^{238}\mathrm{U}\,[e^{\lambda_{238}t} - 1], \qquad {}^{207}\mathrm{Pb}^* = {}^{235}\mathrm{U}\,[e^{\lambda_{235}t} - 1] $$

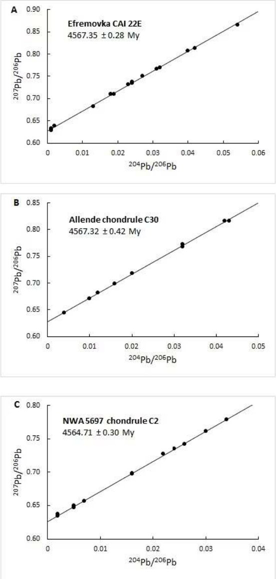
To the board.

Result: The Pb-Pb Isochron

Cogenetic samples with unknown initial Pb fall on a line in ($^{206}\mathrm{Pb}/^{204}\mathrm{Pb}$, $^{207}\mathrm{Pb}/^{204}\mathrm{Pb}$) whose slope depends only on t:

$$\boxed{\frac{^{207}\mathrm{Pb}^*}{^{206}\mathrm{Pb}^*} = \frac{^{235}\mathrm{U}}{^{238}\mathrm{U}}\,\frac{e^{\lambda_{235}t}-1}{e^{\lambda_{238}t}-1}, \qquad \frac{^{235}\mathrm{U}}{^{238}\mathrm{U}} = \frac{1}{137.8}}$$

The slope is independent of the initial Pb and of the U/Pb ratio: the age needs no assumption about the starting composition.



Pb-Pb isochrons for CAI 22E (Efremovka) and chondrules: cogenetic mineral fractions define a linear isochron whose slope yields age *t* independent of initial Pb. From Connelly et al. (2012).

Step 5: Numerical Result

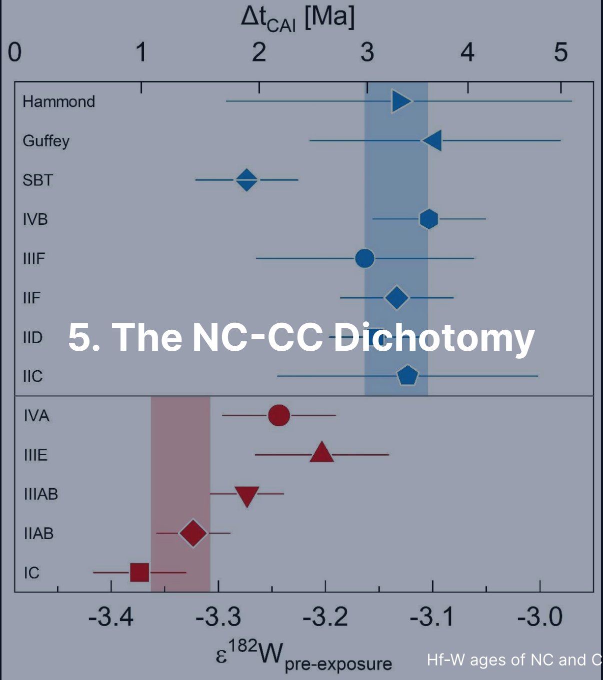
Connelly et al. (2012), CAIs from NWA 2364 (CV3) after acid leaching:

$$\boxed{t_{\mathrm{CAI}} = 4567.30 \pm 0.16\ \mathrm{Myr}}$$

- Independent samples and labs agree to the same precision (Amelin et al. 2010)
- Chondrules from the same meteorite: a few Myr younger
- The *zero point* of the solar system clock

Two parallel decay chains make the problem self-calibrating.

5. The NC-CC Dichotomy



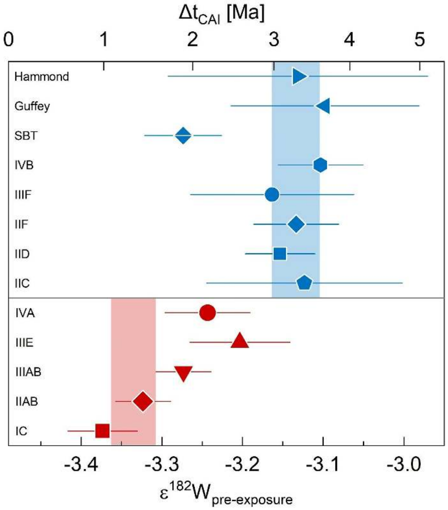
Hf-W ages of NC and CC iron meteorites. Spitzer et al. (2021)

Two Isotopic Reservoirs

Nucleosynthetic anomalies in Ti, Cr, Mo, Ru, and Ca cleanly divide solar system materials into two isolated isotopic reservoirs:

- **NC reservoir (non-carbonaceous):** Ordinary and enstatite chondrites, HEDs, Earth, and Mars
- **CC reservoir (carbonaceous):** Carbonaceous chondrites, several iron groups, and outer bodies
- **Prolonged isolation:** the two reservoirs did not mix isotopically for ~ 3–4 Myr after CAIs

Hf-W Age Separation



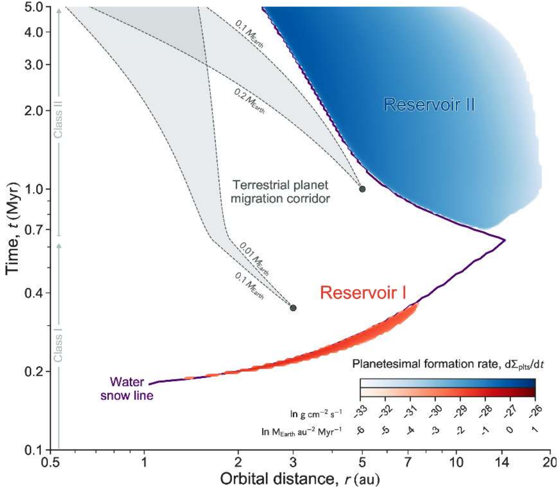
- Two axes, one position: $\varepsilon^{182}\mathrm{W}$ below, the Hf-W age it gives above.
- NC irons (red): −3.37 to −3.20, i.e. Δt_{CAI} = 1.0–2.6 Myr.
- CC irons (blue): −3.16 to −3.10, i.e. 3.0–3.6 Myr.
- Exception: the ungrouped SBT, ~ 1 Myr earlier than the other CC irons.
- NC cores form ~ 2 Myr before CC cores.
- Spitzer et al. (2021).

Three Live Interpretations

Three competing mechanisms explain the NC-CC dichotomy from identical cosmochemical data:

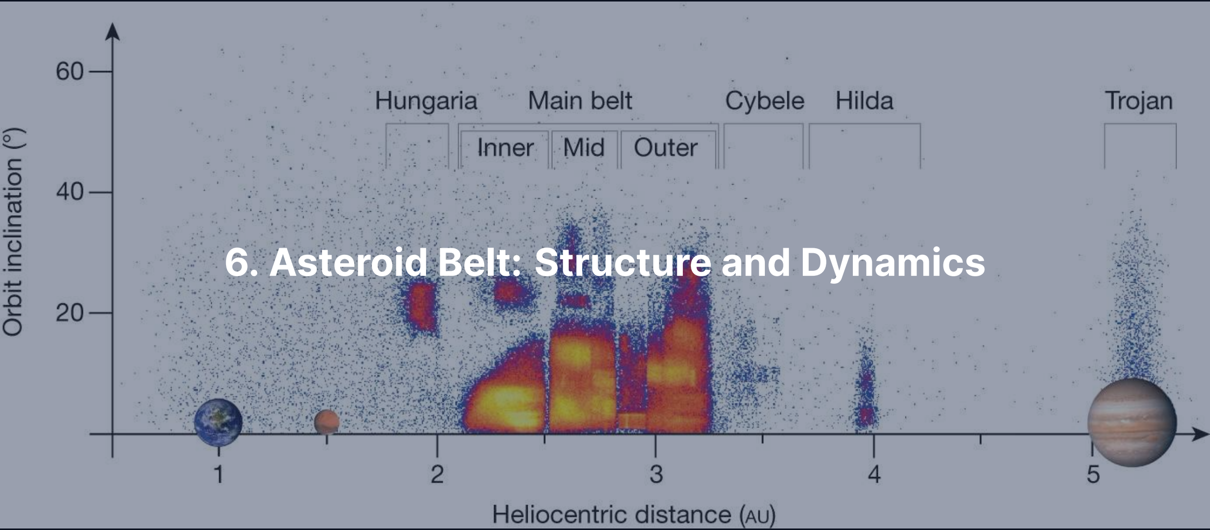
- **Jupiter barrier:** Early proto-Jupiter ($\lesssim$ 1 Myr) opens a gap, blocking inward pebble drift (Kruijer et al. 2017)
- **Snow line:** Disk thermal evolution produces two successive planetesimal formation epochs (Lichtenberg et al. 2021)
- **Temporal change:** Inward disk drift alters feeding composition over time (Schiller et al. 2018)

Snow-Line Mechanism: Two Reservoirs Naturally



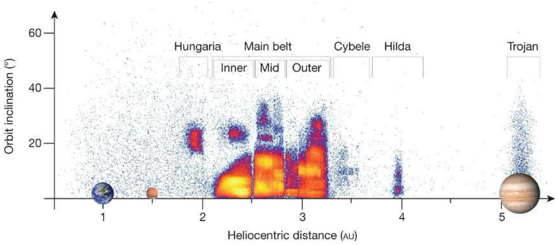
- Disk simulation, time vs. orbital distance.
- Reservoir I (red): early, dry, inner disk = NC.
- Reservoir II (blue): later, ice-rich, outer disk = CC.
- Bifurcation arises from disk evolution itself.
- Lichtenberg et al. (2021).

6. Asteroid Belt: Structure and Dynamics



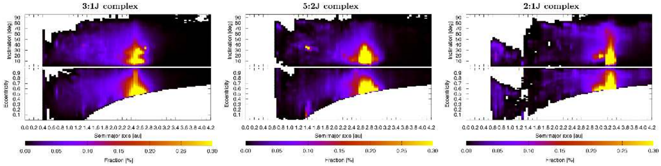
Asteroid orbits in *i* vs *a*. DeMeo & Carry (2014)

The Main Belt in Orbital Space



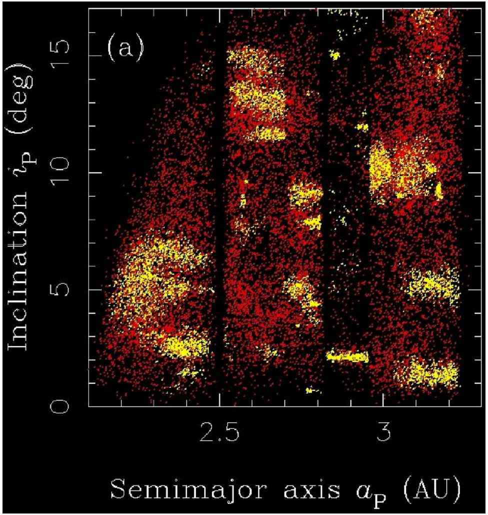
- ~ 1.4 million catalogued bodies; ~ 10^6 at > 1 km.
- Total mass ~ $4 \times 10^{-4}\, M_\oplus$ (Ceres ≈ 40%).
- Hungarias, main belt 2.1–3.3 AU, Hildas, Trojans at 5.2 AU.
- Vertical gaps → Kirkwood resonances with Jupiter.

Kirkwood Gaps as Dynamical Sinks



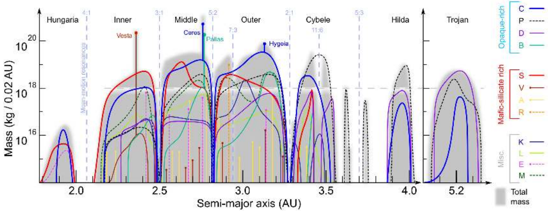
- Strongest resonances: 3:1 (2.50 AU), 5:2 (2.82), 7:3 (2.96), 2:1 (3.27).
- Periodic Jovian kicks add coherently.
- Eccentricity pumped to planet-crossing in 10^4–10^6 yr.
- Gaps are not empty: they are leaky.
- Granvik et al. (2018).

Asteroid Families = Disrupted Parent Bodies



- Hierarchical clustering in proper a, e, i.
- Hirayama families: Themis, Eos, Koronis, Eunomia, Vesta, Flora.
- Each family = debris of a single disruption event.
- Spread → original parent body size, age.

Spectral Classes and the Volatile Gradient



Credit: DeMeo & Carry (2014)

- **S-type** (~ 15% albedo): inner belt; ordinary-chondrite analogues
- **C-type** (~ 4% albedo): ~ 75% of the outer belt; volatile-rich
- **M-type / V-type**: metallic cores and basaltic crust (Vesta family at 2.4 AU)

Radial sorting preserves the disk's primordial volatile gradient.

7. NEAs, Yarkovsky/YORP, and Planetary Defence

Chelyabinsk meteor (2013). A. Alishevskikh CC BY-SA 2.0

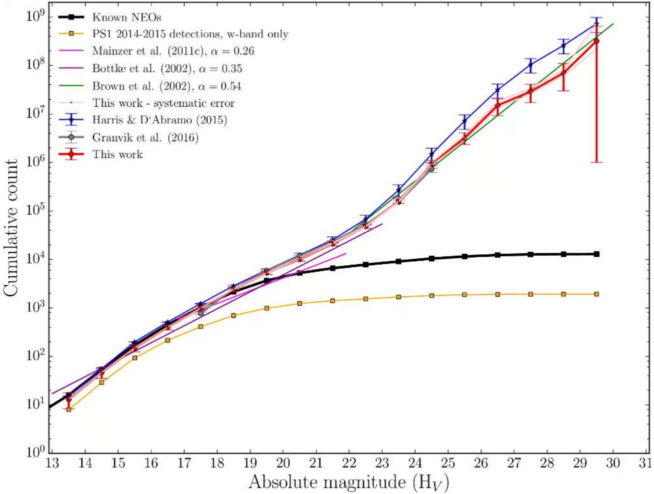
Near-Earth Asteroids and Resupply Channels

NEAs (q < 1.3 AU, residence time ~ 10 Myr) require steady-state resupply from the main belt:

- **Resonances:** Kirkwood (3:1, 5:2) and ν_6 secular resonance pump eccentricity
- **Yarkovsky drift:** Thermal recoil drives slow semimajor axis drift:

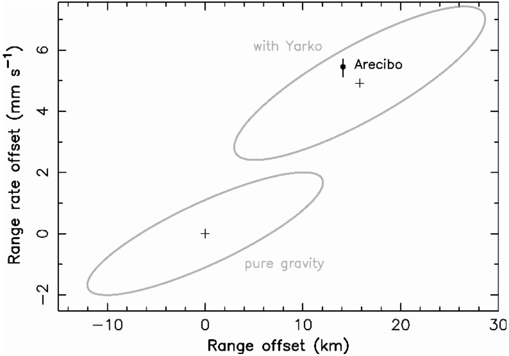
$$\frac{\mathrm{d}a}{\mathrm{d}t} \propto \frac{1}{D\,\rho\,\sqrt{a}}$$

- **Hazards:** ~2,500 Potentially Hazardous Asteroids (D > 140 m, MOID < 0.05 AU)

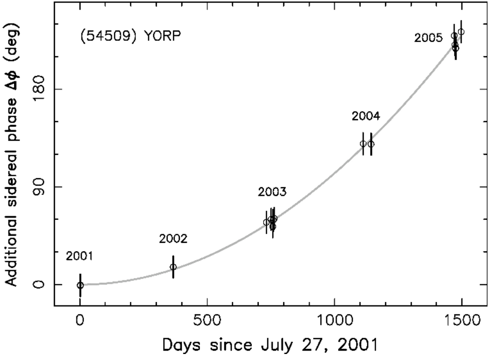


Cumulative size-frequency distribution of near-Earth objects versus absolute magnitude H_V. The power-law distribution spans five orders of magnitude, consistent with steady-state collision and dynamical delivery models. Credit: Schunova et al. (2017), Fig. 9.

Yarkovsky and YORP Detected on Real NEAs



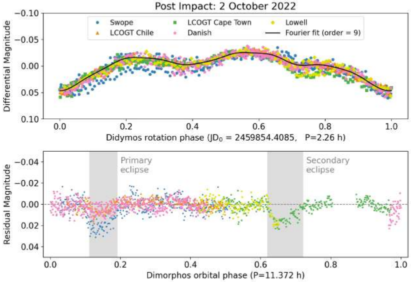
Yarkovsky: (6489) Golevka, Arecibo radar



YORP: (54509) YORP rotational acceleration

YORP: same thermal asymmetry, acting on *spin*. Can spin small asteroids to fission or stop them.

Deflection: DART, the First Practical Test



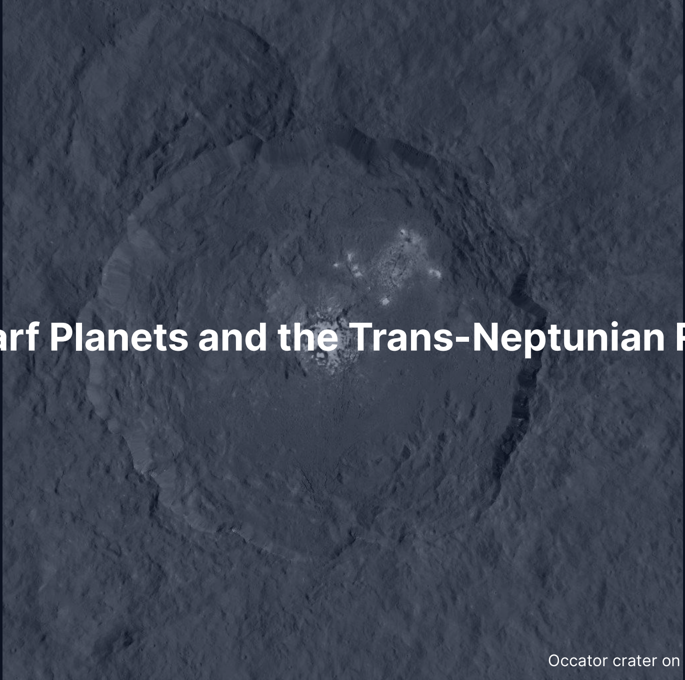
- DART hit Dimorphos (moon of Didymos) on 26 Sep 2022.
- Period reduced by 33.0 ± 1.0 minutes (3σ).
- Momentum enhancement $\beta \approx 3.6$: ejecta recoil exceeds the direct push.
- Hera (ESA, launched 2024) measures the crater and β.
- Rates: 10 m per decade; 50 m (Tunguska) per 10^3 yr; 1 km per 5×10^5 yr; 10 km (Chicxulub) per 10^8 yr

Break

End of Part 1: Meteorites & Inner Small Bodies
Part 2: Trans-Neptunian Region, Comets, Missions

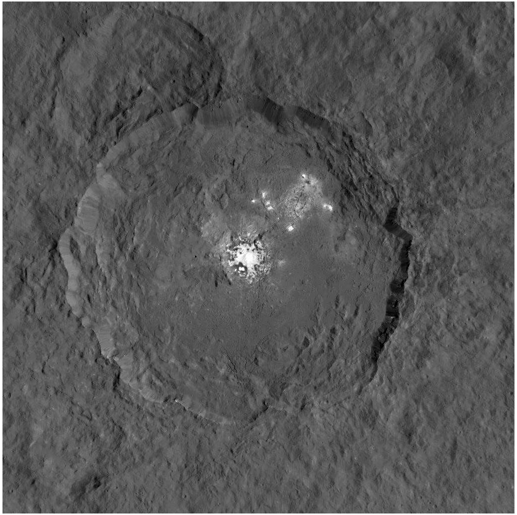
Earthrise, Apollo 8 (1968). Credit: NASA

8. Dwarf Planets and the Trans-Neptunian Region



Occator crater on Ceres, Dawn. NASA/JPL-Caltech

Ceres: Inner-Solar-System Ocean World?



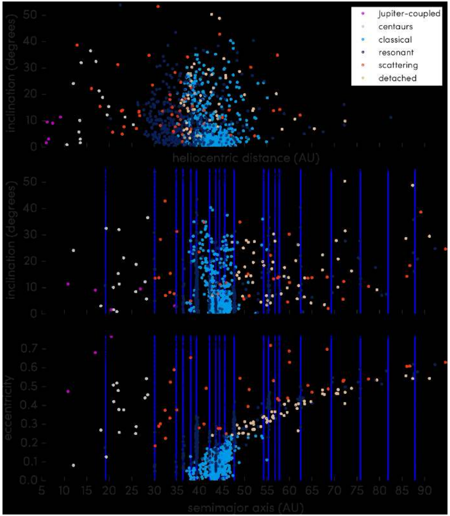
- *R* = 470 km, mass ~ 1% Moon.
- Dawn (2015–2018): hydrated, ammonia-bearing surface.
- Occator: hydrated $\mathrm{Na_2CO_3}$ + $\mathrm{NH_4Cl}$ from erupted brines.
- Implies (or had) a partially liquid water layer at depth.
- Possible captured outer-solar-system body (Nice Model: a dynamical instability among the giant planets).

Trans-Neptunian Region: Five Sub-Populations

The trans-Neptunian region divides into distinct dynamical populations whose orbits preserve the gravitational footprint of giant-planet migration:

- **Classicals (a = 42–47 AU):** Dynamically unperturbed cold classicals ($i \lesssim 5^\circ$) and scattered hot classicals ($i \lesssim 30^\circ$)
- **Resonant (e.g. 3:2 plutinos):** Bodies swept into mean-motion resonances during Neptune's outward migration
- **Scattered disk and detached:** Neptune-scattered bodies ($q \sim 30$ AU) and decoupled detached objects ($q > 40$ AU)

TNO Orbital Architecture (OSSOS)



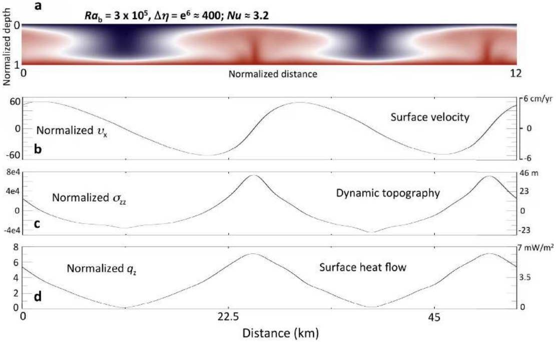
- 1142 characterised TNOs from OSSOS.
- Vertical lines: Neptune resonances.
- Cold classicals: dense low-*i* cluster at 42–47 AU.
- Bannister et al. (2018).
- Architecture is a fossil of giant-planet migration.

9. Pluto, Charon, Arrokoth

Pluto, New Horizons (2015). NASA/JHUAPL/SwRI

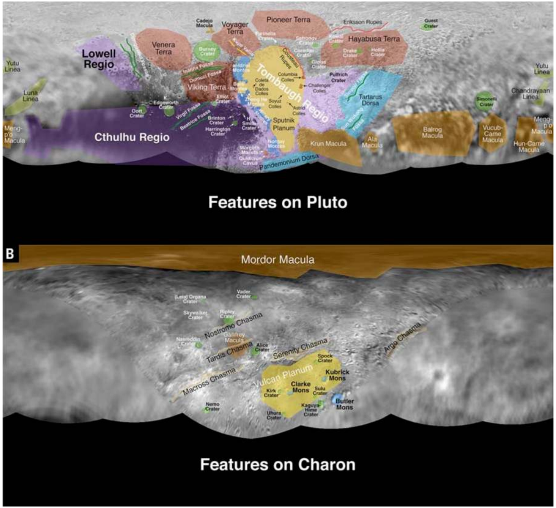
True-colour Pluto from New Horizons Ralph (2015), showing Tombaugh Regio and Sputnik Planitia: with R = 1187 km and ρ = 1860 kg m^{-3}, the world hosts water-ice bedrock overlain by volatile frosts, including a convecting $\mathrm{N_2}$-ice basin younger than 10 Myr; a ~ 1 Pa $\mathrm{N_2}$ atmosphere and a probable subsurface ocean. Credit: NASA/JHUAPL/SwRI.

Solid-State Convection in Sputnik Planitia



- Numerical model, Ra_b ~ 3 × 10^5.
- Cells tens of km across.
- Overturn timescale ~ 10^6 yr.
- Driven by present-day radiogenic heat in the rocky core.
- McKinnon et al. (2016).

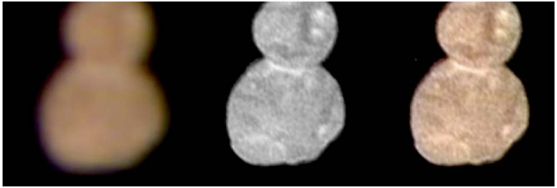
Pluto-Charon: A Binary World



- Charon R = 606 km, ~ 1/8 Pluto's mass.
- Barycentre lies *outside* Pluto.
- Four small moons: Styx, Nix, Kerberos, Hydra.
- Likely fragments of an early collisional event.

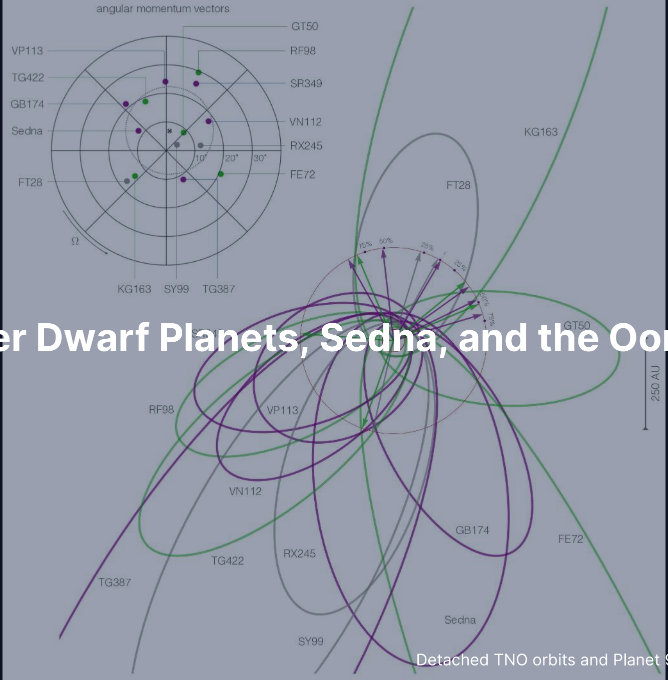
Charon imaged by New Horizons, showing the dark reddish north-polar Mordor Macula formed by photochemically processed tholins sourced from Pluto's escaping atmosphere, alongside equatorial extensional tectonic chasms. Credit: NASA/JHUAPL/SwRI, public domain.

Arrokoth: A Pristine Cold Classical KBO



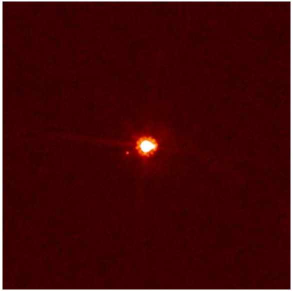
- New Horizons flyby, 1 Jan 2019.
- ~35 km long; **contact binary**.
- Two flat lobes, narrow neck, no impact crater at the join.
- Merger velocity $\lesssim$ few m s^{-1}.
- → formation by streaming-instability gravitational collapse.

10. Other Dwarf Planets, Sedna, and the Oort Cloud

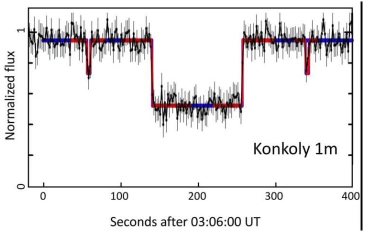


Detached TNO orbits and Planet 9 hypothesis. Batygin et al. (2019)

Eris, Haumea, Makemake



Eris + Dysnomia. More massive than Pluto.

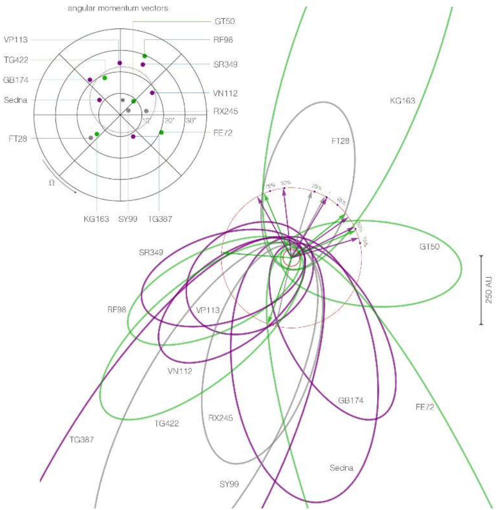


Haumea occultation: revealed a **ring**.

Makemake + small moon. Methane-ice surface.

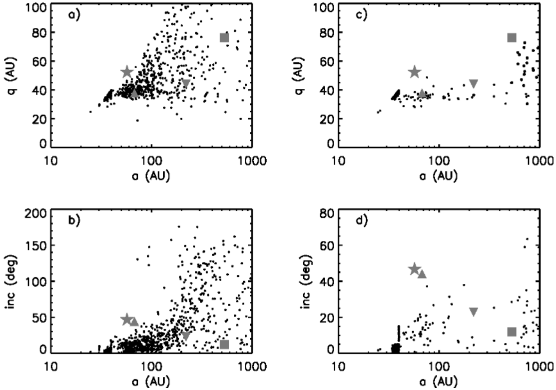
Plus dwarf-planet candidates: Gonggong, Quaoar, Orcus, Salacia, Sedna.

Sedna: Detached, Mysterious



- q = 76 AU, Q = 900 AU. Period 11,400 yr.
- Cannot have been excited by Neptune (q too high).
- Cannot be perturbed by present-day stars (Q too low).
- Possibly inner Oort cloud, or relic of birth-cluster.
- Clustering → **Planet 9** hypothesis (Batygin et al. 2019).

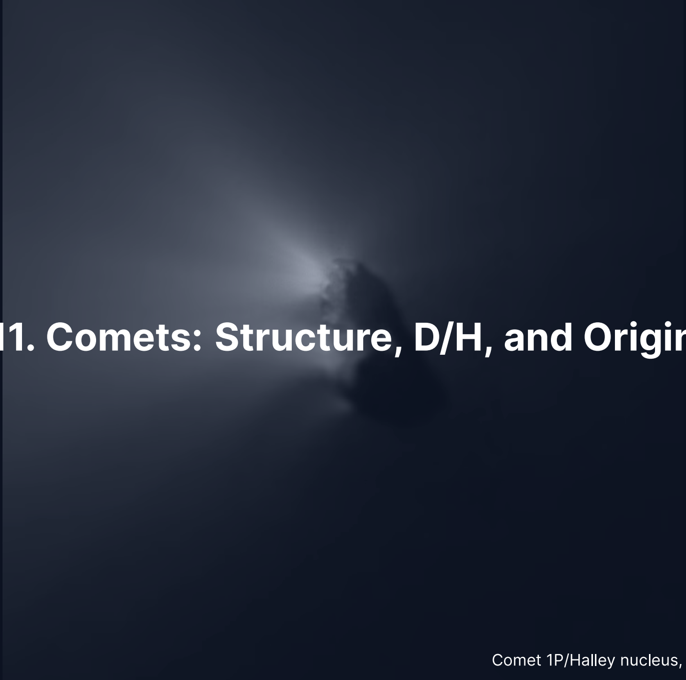
The Oort Cloud



Top row: perihelion q vs semimajor axis a for two simulations of Oort cloud formation. Bottom row: orbital inclination vs a. Kaib & Quinn (2008).

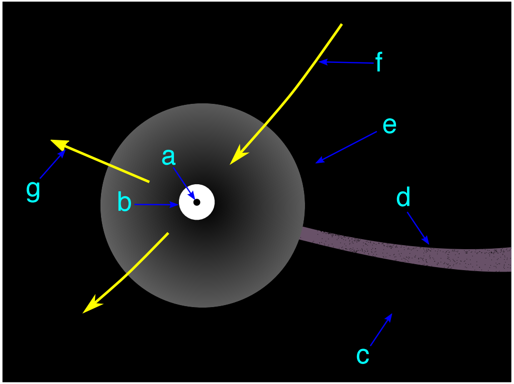
- 2,000–50,000 AU, roughly spherical.
- Inferred 1950 by Oort from long-period comet statistics.
- ~ 10^{11}–10^{12} objects, total mass ~ 1–10 $M_\oplus$.
- Origin: planetesimals scattered out of the Jupiter-Neptune zone, lifted by galactic tide and passing stars.
- **Never directly observed.**

11. Comets: Structure, D/H, and Origin



Comet 1P/Halley nucleus, Giotto (1986). ESA/MPAe Lindau

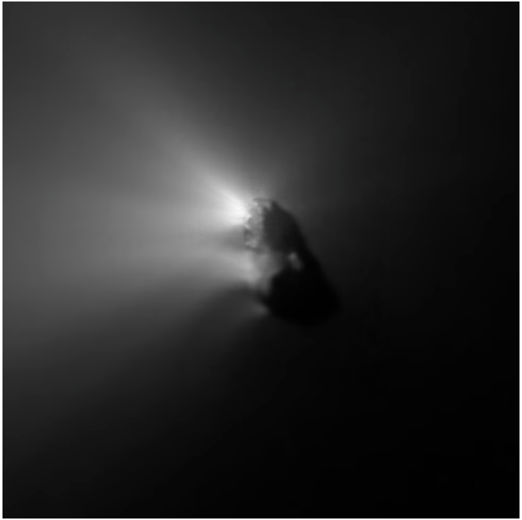
Anatomy of an Active Comet



Credit: Wikimedia Commons, public domain

Whipple's "dirty snowball" (1950), confirmed in detail:

- **Nucleus**: 1–30 km, dark (~ 4% albedo), porous (50–75%), ρ ~ 0.5 g cm^{-3}; 67P (Rosetta): 533 kg m^{-3}, ~ 70% porosity, a contact binary
- Composition: H_2O (dominant), CO, CO_2, CH_4, NH_3, CH_3OH, HCN, dust, organics.
- Inside ~ 5 AU: water sublimation → activity:
  - **Coma**: 10^4–10^6 km gas+dust envelope.
  - **Ion tail**: blue, straight, anti-sunward (solar wind).
  - **Dust tail**: yellow, curved, lags along the orbit.



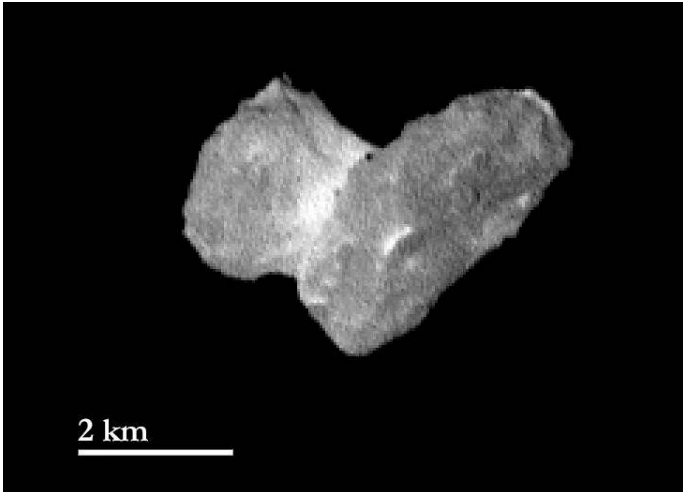
Halley nucleus (15 × 8 × 8 km), Giotto Halley Multicolour Camera, 13–14 March 1986: active jets are confined to small surface fractions of the dark nucleus on its 76-year retrograde orbit ($T_J \approx -0.6$). Credit: ESA/MPAe Lindau.

Short-Period vs Long-Period: The Tisserand Parameter

$$T_J = \frac{a_J}{a} + 2\sqrt{\frac{a}{a_J}(1 - e^2)}\cos i, \qquad a_J = 5.20\,\mathrm{AU}$$

The Tisserand parameter is a conserved dynamical invariant that cleanly separates decoupled asteroids from Jupiter-family and long-period comets:

- **Asteroids ($T_J > 3$):** Main-belt orbits decoupled from Jupiter encounters (e.g. Ceres, $T_J \approx 3.3$)
- **Jupiter-family comets ($2 < T_J < 3$):** Kuiper-belt-derived comets controlled by Jupiter (e.g. 67P, $T_J \approx 2.7$)
- **Long-period comets ($T_J < 2$):** Oort-cloud visitors on high-inclination orbits; Halley-type comets ($T_J \approx -0.6$) are Oort-derived but captured to short orbits (76 yr)



Bilobed nucleus of Jupiter-family comet 67P/Churyumov-Gerasimenko ($T_J \approx 2.7$) imaged by Rosetta OSIRIS, showing two lobes joined at a narrow neck by a gentle primordial contact-binary merger. Credit: ESA/Rosetta/MPS for OSIRIS Team.

D/H of Cometary Water vs Earth

103P/Hartley 2 (EPOXI 2010): D/H ≈ VSMOW

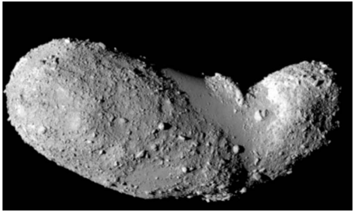
- Earth oceans: D/H = 1.56×10^{-4} (VSMOW standard)
- Hartley 2: ≈ VSMOW; 67P: ≈ 3× VSMOW
- Long-period comets: ≈ 2× VSMOW
- Comets span 3× in D/H: no class match for Earth

Carbonaceous chondrites dominate Earth's water; comets supply at most a minor fraction.

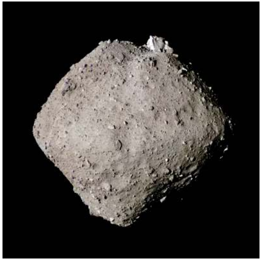
12. Recent Missions and Sample Return

Comet 67P/Churyumov-Gerasimenko, Rosetta. ESA/Rosetta/MPS for OSIRIS Team

Three Sample Returns: Itokawa, Ryugu, Bennu



Hayabusa, 2010: 1,500 grains. Itokawa = LL chondrite parent.



Hayabusa2, 2020: 5.4 g. Ryugu ≈ CI chondrite. Amino acids, nucleobases.

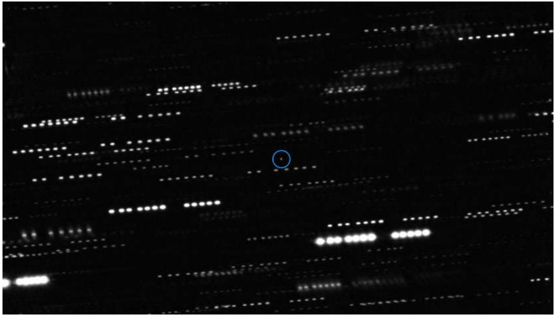
OSIRIS-REx, 2023: 121 g. Bennu CM/CI-like, Mg carbonate veins.

Each rock → specific parent body with a known orbit and full in-situ context.

13. Interstellar Visitors

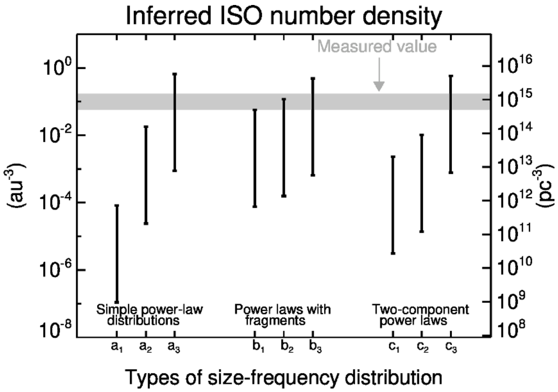
1I/'Oumuamua deep image. ESO / K. Meech et al.

’Oumuamua: First and Strangest

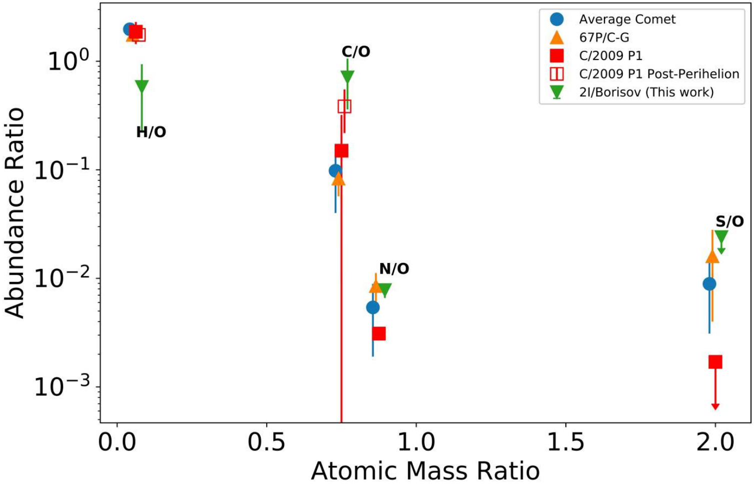


- Centre: ’Oumuamua. Stars trail because the telescope tracked the target.
- No detectable coma at the limit of stacked imaging.
- Yet outgoing trajectory shows non-gravitational acceleration.
- Nature still debated: H_2 ice fragment? N_2 ice? Tidal disruption shard?

Implied ISO Population is Large



- Detection of one ISO with Pan-STARRS implies ~ 10^4 ISOs > 100 m within Neptune's orbit *at any moment*.
- Each ISO is a sample from a different planetary system.
- Cumulative ejection rate: each star ejects planetesimals at rates similar to ours.



Volatile composition in the coma of interstellar comet 2I/Borisov compared with solar-system comets. Elevated C/O and depleted H/O indicate formation beyond the CO snowline of its host planetary system. Credit: Bodewits et al. (2020), Fig. 3.

Summary

1. Small bodies are **formation fossils** preserving disk chemistry and orbital migration
2. **Pb-Pb dating** anchors the solar system's absolute clock at 4567.30 ± 0.16 Myr
3. **NC-CC dichotomy** separates inner and outer reservoirs for ~ 3–4 Myr
4. Resonance gaps and Yarkovsky thermal recoil drive steady-state NEA delivery
5. Cometary D/H shows carbonaceous chondrites delivered most of Earth's water

Next: Lecture 13 turns to the exoplanet revolution.

Questions?

Next: **Lecture 13. Exoplanets**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`